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COURSE NAME: ANTENNA AND WAVE PROPAGATION

EX NO: 10A

ANALYSIS OF CRITICAL FREQUENCY AND MAXIMUM USABLE FREQUENCY (MUF) USING MATLAB

DATE: 11/08/26

OBJECTIVES

i) To simulate the variation of Critical Frequency of the ionosphere using MATLAB and analyze its dependence on electron density.

ii) To determine the Maximum Usable Frequency (MUF) for different angles of incidence and ionospheric conditions using MATLAB.

iii) To compare Critical Frequency and MUF characteristics through graphical analysis and evaluate their impact on long-distance HF communication.

OBJECTIVE 1

ANALYSIS OF CRITICAL FREQUENCY

PROCEDURE

1. Open MATLAB and create a new script.
2. Define the electron density values.
3. Calculate Critical Frequency using the ionospheric model.
4. Plot Critical Frequency versus Electron Density.
5. Analyze the variation of Critical Frequency.

FORMULA

Critical Frequency:

$f^c = 9\sqrt{N_{max}}$

Where:

- f^c = Critical Frequency (MHz)
- N_{max} = Maximum Electron Density (10^{12} electrons/m³)

RESULT :

The Critical Frequency of the ionosphere was successfully calculated and analyzed.

OBJECTIVE 2

DETERMINATION OF MAXIMUM USABLE FREQUENCY (MUF)

PROCEDURE

1. Define the Critical Frequency.
2. Define the angle of incidence.
3. Calculate MUF using the secant law.
4. Evaluate MUF for different angles.
5. Plot MUF variation with angle of incidence.

FORMULA

Maximum Usable Frequency:

MUF = f^c sec θ

Where:

- **f^c** = Critical Frequency
- **θ** = Angle of Incidence

RESULT

The Maximum Usable Frequency for the specified ionospheric conditions was successfully determined.

OBJECTIVE 3

COMPARATIVE ANALYSIS OF CRITICAL FREQUENCY AND MUF

PROCEDURE

1. Calculate Critical Frequency.
2. Calculate MUF for different incidence angles.
3. Plot both parameters on the same graph.
4. Compare propagation characteristics.
5. Analyze communication coverage.

FORMULA :

MUF Ratio:

MUF Ratio = MUF / f^c

RESULT :

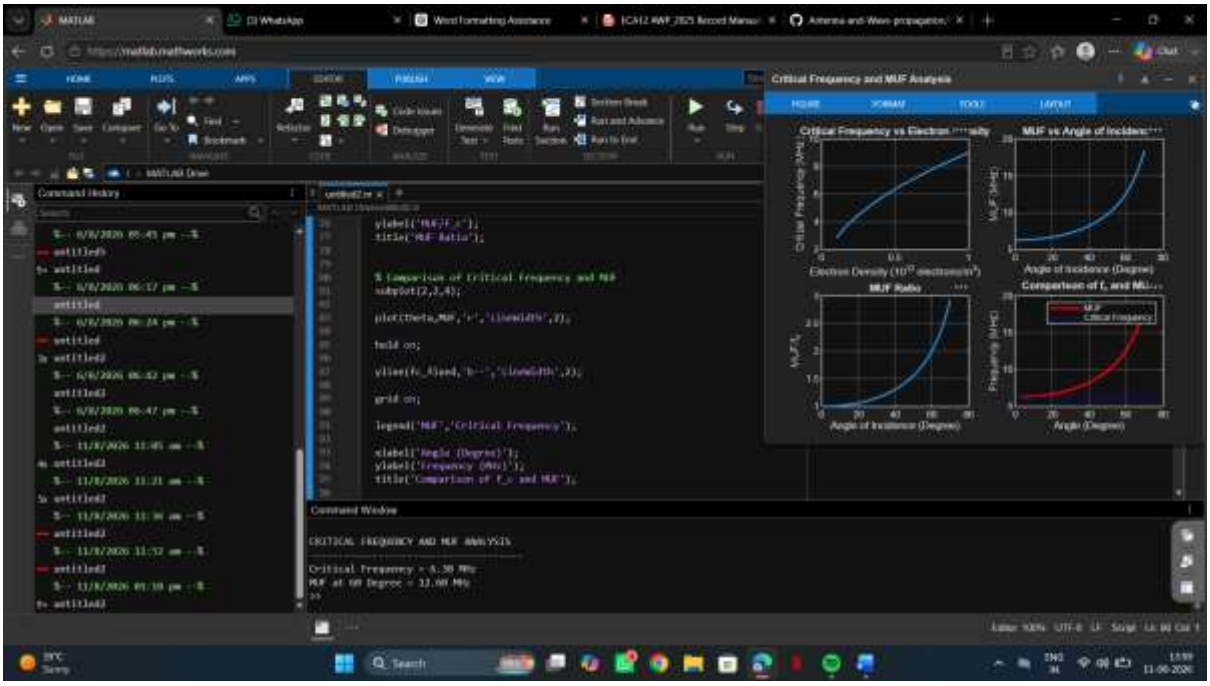
The relationship between Critical Frequency and MUF was successfully analyzed.

OVERALL MATLAB PROGRAM

```
clc;
clear;
close all;
%% Objective 1 : Critical Frequency Analysis
Nmax = 0.1:0.05:1.0;      % Electron Density (10^12 electrons/m^3)
fc = 9*sqrt(Nmax);        % Critical Frequency (MHz)
%% Objective 2 : MUF Analysis
theta = 0:1:70;           % Angle of Incidence (degrees)
fc_fixed = 6.3;           % Fixed Critical Frequency (MHz)
MUF = fc_fixed*sec(deg2rad(theta));
%% Objective 3 : Comparative Analysis
MUF_ratio = MUF/fc_fixed;
%% Display Results
fprintf('\nCRITICAL FREQUENCY AND MUF ANALYSIS\n');
fprintf('-------------------------------------\n');
fprintf('Critical Frequency = %.2f MHz\n', fc_fixed);
fprintf('MUF at 60 Degree = %.2f MHz\n', ...
   fc_fixed*sec(deg2rad(60)));
%% Plot Results
figure('Name', ...
      'Critical Frequency and MUF Analysis', ...
      'NumberTitle','off');
```

```
% Critical Frequency vs Electron Density

subplot(2,2,1);

plot(Nmax,fc,'LineWidth',2);

grid on;

xlabel('Electron Density (10^{12} electrons/m^3)');

ylabel('Critical Frequency (MHz)');

title('Critical Frequency vs Electron Density');

% MUF vs Angle of Incidence

subplot(2,2,2);

plot(theta,MUF,'LineWidth',2);

grid on;

xlabel('Angle of Incidence (Degree)');

ylabel('MUF (MHz)');

title('MUF vs Angle of Incidence');

% MUF Ratio

subplot(2,2,3);

plot(theta,MUF_ratio,'LineWidth',2);

grid on;

xlabel('Angle of Incidence (Degree)');

ylabel('MUF/f_c');

title('MUF Ratio');

% Comparison of Critical Frequency and MUF

subplot(2,2,4);

plot(theta,MUF,'r','LineWidth',2);

hold on;

yline(fc_fixed,'b--','LineWidth',2);

grid on;

legend('MUF','Critical Frequency');

xlabel('Angle (Degree)');

ylabel('Frequency (MHz)');

title('Comparison of f_c and MUF');
```

OUTPUT :



FINAL RESULT

The Critical Frequency and Maximum Usable Frequency (MUF) were successfully analyzed using MATLAB. The variation of Critical Frequency with electron density and the variation of MUF with angle of incidence were plotted and studied. The relationship between Critical Frequency and MUF was also successfully analyzed.